2.4 Pre-Quiz: Lines, Exponents and Radicals — Markscheme

Total: 77 marks.

1. [Maximum mark: 14]

(a) $m_{AB} = \dfrac{5-1}{6-(-2)}$ **M1**

$m_{AB} = \dfrac{4}{8} = \dfrac{1}{2}$ **A1**

A bare answer earns M0A1.

(b) $m_L = -2$ **A1**

$y-(-2) = -2(x-7)$ **M1**

$y = -2x+12$ **A1**

A1 for the negative reciprocal of their gradient. M1 for substituting their gradient and $C(7,-2)$ into any form of the equation of a line. The final A1 needs the form $y = mx+c$. FT from (a).

Wrong method: the perpendicular bisector of AB, $y=-2x+7$, passes through the midpoint of AB, not through C: A1M0A0.

(c) line AB: $y-1 = \frac{1}{2}(x+2)$, so $y = \frac{1}{2}x+2$ **A1**

$-2x+12 = \frac{1}{2}x+2$ **M1**

$\frac{5}{2}x = 10$

$x=4, \quad y=4 \qquad D(4,\ 4)$ **A1A1**

M1 for equating their two lines, or for any valid method of solving them together (substitution, elimination, GDC). A1 for each coordinate. FT from (b). Wrong method: the midpoint of AB, $(2,\ 3)$, is not the foot of the perpendicular from C: A1 for the equation of AB if seen, then M0A0A0.

(d) $CD = \sqrt{(7-4)^2+(-2-4)^2} = \sqrt{45}$ **M1**

$CD = 3\sqrt{5}$ **A1**

Accept $a=3$. $\sqrt{45}$ or 6.71 alone earns A0. FT from (c). Wrong method: $CA = 3\sqrt{10}$ or $CB = 5\sqrt{2}$ is not the perpendicular distance: M0A0.

(e) $AB = \sqrt{8^2+4^2} = \sqrt{80} = 4\sqrt{5}$ **A1**

Area $= \frac{1}{2}\left(4\sqrt{5}\right)\left(3\sqrt{5}\right)$ **M1**

Area $= 30$ **A1**

M1 for $\frac{1}{2} \times$ base $\times$ height with AB as the base and CD as the height. Accept any other correct method, e.g. enclosing the triangle in a rectangle and subtracting. FT from (d).

2. [Maximum mark: 10]

(a) $m = \dfrac{305 - 165}{7 - 3}$ **M1**

$m = 35$ **A1**

(b) $165 = 35(3) + c$ **M1**

$c = 60$ **A1**

Accept solving the system $3m + c = 165$, $7m + c = 305$ *for both values at once. FT from (a).*

(c) $c = 60$ is the fixed fee: company A charges 60 euros before any days are added. **A1**

Accept "the cost for 0 *days" or "the fixed charge". Do not accept "the y-intercept" with no reference to the context.*

(d) $35d + 60 = 45d + 10$ **M1**

$10d = 50$

$d = 5$ **A1**

Accept a graphical or table method on the GDC. FT from (a) and (b).

(e) $C_A(9) = 35(9) + 60 = 375, \quad C_B(9) = 45(9) + 10 = 415$ **M1**

company A is cheaper **A1**

she saves $415 - 375 = 40$ euros **A1**

M1 for evaluating both costs at $d = 9$. *FT from (a) and (b).*

3. [Maximum mark: 13]

(a) $3x - 4y = 18$ and $10x + 4y = 8$, adding: $13x = 26$ **(M1)**

$x = 2$ **A1**

$y = -3$ **A1**

M1 for any valid method: elimination, substitution, or the GDC simultaneous-equation solver. A correct answer from the GDC with no other working earns full marks.

(b) L_2: $y = \frac{1}{2}x - \frac{5}{6}$, gradient $\frac{1}{2}$ **A1**

L_1: $y = -\frac{2}{k}x + \frac{6}{k}$, so $-\frac{2}{k} = \frac{1}{2}$ **M1**

$k = -4$ **A1**

M1 for equating the gradients of the two lines. Accept comparing coefficients: $\frac{2}{3} = \frac{k}{-6}$.

(c) no solutions **A1**

the lines are parallel and are not the same line (y-intercepts $-\frac{3}{2}$ and $-\frac{5}{6}$), so they never meet **R1**

Accept "parallel lines do not intersect". Do not award A1R0.

(d) $\left(-\frac{2}{k}\right) \times \frac{1}{2} = -1$ **M1**

$k = 1$ **A1**

M1 for using the product of the gradients $= -1$*, or the gradient of* L_1 *equal to* -2*. FT from the gradient of* L_2 *in (b).*

(e) $2x + y = 6, \quad 3x - 6y = 5$

$3x - 6(6 - 2x) = 5$ **M1**

$15x = 41, \quad x = \dfrac{41}{15}$ **A1**

$y = 6 - 2\left(\dfrac{41}{15}\right) = \dfrac{8}{15}$ **A1**

M1 for any valid method, including the GDC solver. The question asks for exact values: decimal coordinates $(2.73,\ 0.533)$ *earn A1A0. FT from (d).*

4. [Maximum mark: 15]

(a) $(2x^3y^{-2})^3 = 8x^9y^{-6}$ **M1**

$\dfrac{8x^9y^{-6}}{4x^5y^{-4}} = 2x^4y^{-2}$ **A1**

$= \dfrac{2x^4}{y^2}$ **A1**

M1 for raising every factor of the bracket to the power 3. *The final A1 needs positive exponents only.*

(b) $27^{-\frac{2}{3}} = \dfrac{1}{\left(\sqrt[3]{27}\right)^2} = \dfrac{1}{9}$ **A1**

$(x^6)^{-\frac{2}{3}} = x^{-4}$, so the expression is $\dfrac{1}{9}x^{-4}$ **A1**

Accept $a = \frac{1}{9}$, $n = -4$. *Do not accept* $0.111x^{-4}$*: a must be exact.*

(c) $(3^2)^{x+1} = (3^{-3})^x$ **M1**

$2x + 2 = -3x$ **A1**

$x = -\dfrac{2}{5}$ **A1**

M1 for writing both sides as powers of 3. *Accept* $x = -0.4$. *A GDC answer with no working earns M0A0A1.*

(d) $4^x = (2^2)^x = (2^x)^2 = u^2$ **A1**

so $2u^2 - 17u + 8 = 0$ **AG**

(e) $(2u - 1)(u - 8) = 0$ **M1**

$u = \frac{1}{2}$ or $u = 8$ **A1**

$2^x = \frac{1}{2} \Rightarrow x = -1$ **A1**

$2^x = 8 \Rightarrow x = 3$ **A1**

M1 for solving the quadratic in u *by any method (factorizing, formula, GDC). Stopping at* $u = \frac{1}{2}, 8$ *earns M1A1A0A0. Reading* 2^x *as* $2x$ *gives* $x = \frac{1}{4}, 4$*: A0A0 for those values. "Hence" requires the substitution: reading both roots from a GDC graph of the original equation, with no work in* u*, earns M0A0A1A1.*

(f) graph $y = 5^x$ and $y = 40$ and find the intersection **(M1)**

$x = 2.29 \quad (2.29203\ldots)$ **A1**

The M1 is implied by a correct answer, so a bare 2.29 *earns 2/2. A1 for* 2.29 *or the display with an ellipsis;* 2.292 *with no ellipsis earns A0.*

5. [Maximum mark: 13]

(a) $5\sqrt{3}+4\sqrt{3}-6\sqrt{3}$ **M1**

$=3\sqrt{3}$ **A1**

M1 for writing at least two of the three surds as multiples of $\sqrt{3}$.

(b) $\left(2\sqrt{5}\right)^2-2\left(2\sqrt{5}\right)(3)+3^2=20-12\sqrt{5}+9$ **M1**

$=29-12\sqrt{5}$ **A1**

M1 for an expansion with all three (or four) terms.

(c) $\dfrac{\sqrt{3}+1}{\sqrt{3}-1}\times\dfrac{\sqrt{3}+1}{\sqrt{3}+1}$ **M1**

$=\dfrac{3+2\sqrt{3}+1}{3-1}=\dfrac{4+2\sqrt{3}}{2}$ **A1**

$=2+\sqrt{3}$ **A1**

M1 for multiplying numerator and denominator by the conjugate $\sqrt{3}+1$. *A1 for a correct numerator and the denominator* 2. *Accept* $a=2$, $b=1$.

(d) $x\sqrt{8}-x\sqrt{2}=8$ **M1**

$2x\sqrt{2}-x\sqrt{2}=8, \quad \text{so } x\sqrt{2}=8$ **A1**

$x=\dfrac{8}{\sqrt{2}}=\dfrac{8\sqrt{2}}{2}=4\sqrt{2}$ **A1**

M1 for collecting the x*-terms on one side. A1 for* $x\sqrt{2}=8$, *or* $x\left(\sqrt{8}-\sqrt{2}\right)=8$ *with* $\sqrt{8}=2\sqrt{2}$ *seen. The final A1 needs the form* $a\sqrt{2}$; $\frac{8}{\sqrt{2}}$ *or* 5.66 *earns A0.*

(e) width $=\dfrac{7+7\sqrt{2}}{3+\sqrt{2}}\times\dfrac{3-\sqrt{2}}{3-\sqrt{2}}$ **M1**

$=\dfrac{21-7\sqrt{2}+21\sqrt{2}-14}{9-2}=\dfrac{7+14\sqrt{2}}{7}$ **A1**

$=1+2\sqrt{2}$ **A1**

Accept the system method: $(3+\sqrt{2})(a+b\sqrt{2})=7+7\sqrt{2}$ *gives* $3a+2b=7$ *and* $a+3b=7$ *(M1A1), so* $a=1$, $b=2$ *(A1).*

6. [Maximum mark: 12]

(a) $m = \dfrac{7-3}{3\sqrt{2}-\sqrt{2}}$ **M1**

$m = \dfrac{4}{2\sqrt{2}} = \dfrac{2}{\sqrt{2}} = \dfrac{2\sqrt{2}}{2}$ **A1**

$m = \sqrt{2}$ **AG**

A1 for a correct step that removes the surd from the denominator, or for $\frac{2}{\sqrt{2}} = \frac{\sqrt{2}\sqrt{2}}{\sqrt{2}}$. *A decimal check (*$1.414\ldots$*) does not show the result: A0.*

(b) $y - 3 = \sqrt{2}\left(x - \sqrt{2}\right)$ **M1**

$y = \sqrt{2}x - 2 + 3 = \sqrt{2}x + 1$ **A1**

M1 for substituting $\sqrt{2}$ *and either point into any form of the equation of a line. The A1 needs* $\sqrt{2} \times \sqrt{2} = 2$ *and the form* $y = mx + c$.

(c) $0 = \sqrt{2}x + 1$ **M1**

$x = -\dfrac{1}{\sqrt{2}} = -\dfrac{\sqrt{2}}{2}$ **A1**

Accept $\left(-\frac{\sqrt{2}}{2},\ 0\right)$. $-\frac{1}{\sqrt{2}}$ *or* -0.707 *earns A0. FT from (b).*

(d) $\sqrt{2}x + 1 = 2x, \quad x\left(2 - \sqrt{2}\right) = 1$ **M1**

$x = \dfrac{1}{2-\sqrt{2}} \times \dfrac{2+\sqrt{2}}{2+\sqrt{2}} = \dfrac{2+\sqrt{2}}{4-2}$ **M1**

$x = 1 + \dfrac{1}{2}\sqrt{2}$ **A1**

$y = 2x = 2 + \sqrt{2}$ **A1**

First M1 for equating the two lines and collecting the x*-terms. Second M1 for multiplying by the conjugate* $2 + \sqrt{2}$. *Accept* $x = \frac{2+\sqrt{2}}{2}$. *Decimal coordinates* $(1.71,\ 3.41)$ *earn M1M0A0A0. FT from (b).*

(e) $AB = \sqrt{\left(3\sqrt{2} - \sqrt{2}\right)^2 + (7-3)^2} = \sqrt{8+16}$ **M1**

$AB = \sqrt{24} = 2\sqrt{6}$ **A1**

The A1 needs simplest surd form: $\sqrt{24}$ *or* 4.90 *earns A0.*